

Alejandro Rodas

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EDUCATION

•University of California, Berkeley (CDSS)

2021 – 2025

Bachelor of Arts in Data Science (Industrial Analytics Emphasis)

Berkeley, CA

Relevant Coursework: Data Engineering, Data Mining & Analytics, Probability for Data Science, Data Inference & Decisions, Principles & Techniques of Data Science, Data Structures & Algorithms.

Awards & Scholarships: Generation Change Scholar, James Hjul Scholar, Allmond Scholar, Albert Job Scholar.

TECHNICAL SKILLS

Skills/Languages: Advanced Python, Java, Advanced SQL, MongoDB, GCP, AWS, ETL, Databases, Git, Cloud, ETL Pipelines, API experience, Relational Databases, Kafka, data modeling, CI/CD workflows, Pyspark/Spark, Snowflake, BigQuery, RAG Pipelines, LLM orchestration pipelines

EXPERIENCE

•UC Berkeley Financial Aid & Scholarships

Mar 2022 – May 2025

Data Science Intern

Berkeley, CA

- Designed Python and SQL-based **data pipelines** and monitoring tools to track usage of scholar spaces.
- Cleaned and integrated multi-year student and aid datasets into **relational schemas**, optimizing **SQL** queries for large-scale reporting.
- Designed and analyzed **A/B tests** on outreach emails and student portals, improving FAFSA completion and critical financial-aid form response rates by **12%**.
- Built **GCP pipelines** from S3 datasets on aid usage, unmet need, retention, and first-gen/low-income outcomes to support policy decisions by financial officers.

•U.S. Health Department (HHS)

May 2024 – Aug 2024

Data Science Intern

Washington, D.C. (Remote)

- Developed statistical/ML models and **A/B tests** on CIL (U.S. Health Dept.Centers) funding and utilization data (2019 vs. 2022) to quantify post-pandemic underperformance.
- Helped to automate **ETL pipelines** using Python and APIs, reducing data processing time by **30%**.
- Leveraged **AWS pipelines** and Tableau dashboards summarizing funding and usage trends for non-technical decision-makers.
- Synthesized pre-pandemic and post-pandemic funding and utilization findings with executive directors, contributing statistical analyses and visualizations to a federal report scheduled for release in 2026.

•CDSS Discovery Research Program

Jan 2024 – May 2024

Lead Researcher

Berkeley, CA

- **Led a research team analyzing federal reports for 350+** CIL (U.S. Health Dept.Centers), cleaning and standardizing multi-year CSVs into a state-level dataset.
- Helped analyzing and developing **cloud infrastructure and ingestion pipelines** for future analysis/automation of processes within the organization.
- Merged CIL data with U.S. Census disability statistics to engineer coverage and funding-per-disabled-resident metrics capturing “good” CIL performance. Finally presented findings at the 2024 CDSS Data Discovery Spring Symposium to inform discussions on HHS resource allocation and to our policy team based in Princeton (Mathematica, Inc.)

HIGHLIGHTED PROJECTS

•Campus Sensor Data Cleaning and Time-Series Interpolation in PostgreSQL

Fall 2025

- Standardized HVAC/ energy sensor data in **PostgreSQL**, resolving inconsistent units and building metadata tables.
- Engineered 15-minute analysis-ready time-series tables using **SQL window functions**, and outlier detection based on statistical models to fill missing readings.
- Produced clean, structured datasets used for **exploratory analysis** of campus energy usage.

•Yelp Insights and NoSQL Data Processing with MongoDB

Fall 2025

- Queried and analyzed semi-structured JSON data from Yelp business, review, and user datasets using PyMongo.
- Designed **data pipelines and ETL pipelines** for aggregating and extracting user behavior insights.

•AI Agents for Campaign Analytics

Spring 2025

- Developed an AI-powered campaign analytics agent using **LangChain**, **OpenAI APIs**, and **BigQuery** to enable natural language querying and retrieval over marketing campaign data.
- Built a **RAG pipeline** and ETL pipelines to aggregate, visualize, and infer ROI analytics within a graph-based multi-agent workflow, generating automated insights and charts for campaign performance evaluation.

•Causal Inference and ML (Python) for U.S. Primary Elections

Spring 2025

- Conducted **causal analysis and Bayesian statistics** (e.g. Propensity Score Matching) to estimate endorsement effects.
- Built and evaluated logistic regression and Random Forest models; **performed EDA on 2,600+ candidates**.
- Tuned models; best Random Forest achieved **85% accuracy** and 0.78 F1-score; presented findings using **Matplotlib and Seaborn** in a 23-page paper.